

Evaluating Social Bias in Large Language Models with BBQ

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Project repository: github.com/pminhtaiiii/LLM-Bias-Research

Abstract

This report evaluates ten large language model (LLM) conditions on all 58,492 question-answering instances in the Bias Benchmark for Question Answering (BBQ). BBQ separates ambiguous contexts, where the correct response is an uncertainty option, from disambiguated contexts, where the evidence supports a specific answer.

The report uses ambiguous and disambiguated accuracy together with the directional bias scores s_{AMB} and s_{DIS} to assess whether model answers tend to follow stereotypes. It also uses the evidence–stereotype alignment gap to examine whether model accuracy decreases when the correct answer conflicts with the stereotype.

The results show that ambiguous- and disambiguated-context performance forms a major axis of observed variation across evaluation conditions. Granite 4.1 3B reaches 94.27% ambiguous accuracy, while Phi-4 Mini and Ministral reach 66.34% and 18.25%, even though the latter two exceed 92% disambiguated accuracy. Physical Appearance and Age show recurring larger observed ambiguous-context bias scores for the local models. Qwen3 4B shows a smaller AMB–DIS accuracy disparity than Phi-4 Mini and Ministral, but contains 49 strict-format invalid outputs. Several GPT medium/light conditions show the opposite pattern: strong ambiguous accuracy but weak disambiguated accuracy, while their directional bias scores remain close to zero.

Two GPT High-condition answer sequences are identical and perfectly match the gold answers. Because the full inference configuration and server-side processing details were not available, the original inference process could not be independently reproduced, and the reason for the identical answer sequences and 100% accuracy could not be determined with certainty. Therefore, these results are reported but are not used as the main basis for comparing model performance.

1 Introduction

Large language models can still produce responses that reflect social stereotypes, even when they perform well on conventional NLP benchmarks.

In question answering, there are two main problems. First, when the available information is insufficient, a model may make a guess based on stereotypes. Second, even when clear evidence is provided, stereotypes may still influence the model’s answer.

The BBQ benchmark was created to examine these two problems using multiple-choice questions (Parrish et al., 2022). This study uses BBQ to evaluate ten conditions across six base model labels.

The study focuses on three research questions:

1. **RQ1:** How does model performance differ between ambiguous and disambiguated BBQ contexts?
2. **RQ2:** How strongly do model responses align with the stereotype direction measured by BBQ under ambiguous and disambiguated contexts?
3. **RQ3:** Does accuracy change when disambiguating evidence conflicts with the stereotype-aligned target?

All 58,492 BBQ instances are used to evaluate the models based on accuracy, bias scores, results across different categories, and data quality.

An important point is that a bias score close to zero does not necessarily mean that a model performs well. A model may still perform poorly when the information is insufficient or fail

to follow the available evidence correctly.

2 Related Work

Measuring Bias in NLP. BBQ frames social-bias evaluation as observable behavior in a controlled question-answering setting, rather than treating a bias score alone as a complete description of model behavior (Parrish et al., 2022).

Social Biases in Question Answering. BBQ uses the answers that models actually choose in QA tasks to examine whether models rely on stereotypes when information is insufficient and whether stereotypes still influence their answers even when clear evidence is available (Parrish et al., 2022).

Evaluating Social Bias in LLMs with BBQ. This report builds on the BBQ benchmark to evaluate ten evaluation conditions across six base model labels on the full BBQ evaluation set. The analysis focuses on comparing model performance on ambiguous and disambiguated questions, measuring directional bias using s_{AMB} and s_{DIS} , and examining whether model accuracy decreases when the correct answer conflicts with a social stereotype. The study also compares bias patterns across categories and analyzes differences between local and cloud-based models.

3 BBQ Benchmark

The evaluation uses all 58,492 BBQ question-answering instances. Each instance presents two people or groups plus an uncertainty answer.

The benchmark combines two context conditions with two question polarities.

3.1 Context Conditions

Ambiguous contexts. The context is intentionally under-informative, so the gold answer is UNKNOWN.

Disambiguated contexts. The context provides enough evidence to identify the correct answer.

3.2 Question Polarity

Questions are either **negative** or **non-negative**.

Negative questions ask about a stereotyped property, while non-negative questions ask about its complement.

The public BBQ data contain 11 source categories. The reporting taxonomy used in this study splits Gender Identity and Race/Ethnicity into explicit-label and proper-name variants, yielding 13 analysis categories.

3.3 Evaluation Metrics

A response is treated as stereotype-aligned when the model selects the bias target for a negative question or the corresponding non-target for a non-negative question. The opposite response direction is treated as anti-biased. Thus, the direction of bias depends jointly on question polarity and target/non-target metadata.

For valid, target-scorable ambiguous rows,

$$s_{AMB} = \frac{n_{biased} - n_{anti}}{N_{valid, target}}. \quad (1)$$

UNKNOWN responses remain in the denominator and contribute zero.

For disambiguated rows,

$$s_{\text{DIS}} = 2 \frac{n_{\text{biased}}}{n_{\text{biased}} + n_{\text{anti}}} - 1. \quad (2)$$

Positive values indicate a greater tendency toward stereotype-aligned responses, while negative values indicate a greater tendency toward anti-biased responses.

The report also uses

$$\Delta_{\text{align}} = \text{Acc}_{\text{nonaligned}} - \text{Acc}_{\text{aligned}}. \quad (3)$$

A negative value indicates lower disambiguated accuracy when the evidence-supported answer conflicts with the stereotype-aligned target.

4 Experimental Setup

4.1 Models and Inference Conditions

Local-model protocol. Granite 4.1 3B, Phi-4 Mini 3.8B, and Ministral 3 3B used the same deterministic local multiple-choice protocol with temperature = 0, top-p = 1, seed = 1234, maximum output length = 384, context length = 2048, Q4_K_M GGUF files, and the llama.cpp runtime.

The prompt instructed the models to use only the provided context and return exactly one answer: A, B, or C.

Qwen3 4B protocol. Qwen3 4B Instruct used a Q6_K GGUF file on Kaggle with llama-cpp-python 0.3.35, temperature = 0, context length = 2048, maximum output length = 10, and /no_think. Its parser accepted only an exact A/B/C after removing any thinking block and normalizing whitespace and case.

Cloud-model conditions. GPT 5.5 and GPT 5.6 Luna were evaluated under High, Medium, and Light conditions. Complete answer sequences were available, but the labels and server-side configurations are retained from supplied artifacts and are not fully documented as directly comparable inference settings.

4.2 Scoring and Data Quality

All outputs were aligned to a canonical 58,492-row benchmark index.

Sixteen rows do not have usable target_loc; they remain in accuracy calculations but are excluded from directional bias and alignment metrics.

Qwen3 4B produced 49 strict-format invalid outputs. Under our strict protocol-compliance rule, these outputs were not rescued post-hoc and were counted as incorrect for accuracy evaluation.

An earlier Qwen aggregate contained 58,556 rows because of duplicated metadata; the canonical analysis uses the 58,492 unique raw predictions.

The GPT 5.5-High and GPT 5.6-Luna-High answer sequences are identical and perfectly match the benchmark gold answers.

These results are retained but are not used as the main basis for model comparison.

The accompanying [project repository](#) provides the available prompts, scoring scripts, model metadata, benchmark lock, and derived result tables.

5 Results

5.1 Overall Accuracy

Table 1 reports the headline metrics.

Note. * denotes audit-limited GPT High conditions whose supplied answer sequences are identical and perfectly match the benchmark gold answers. Qwen3 4B contains 49 strict-format invalid outputs.

As shown in Figure 1, the three local models differ most strongly in ambiguous-context behavior.

Granite 4.1 3B reaches 94.27% AMB accuracy, compared with 66.34% for Phi-4 Mini and 18.25% for Ministral 3 3B.

Their DIS accuracies are much closer: 87.71%, 93.67%, and 92.81%, respectively.

Ministral therefore follows explicit evidence well but frequently fails to select UNKNOWN when the context is under-informative.

Qwen3 4B shows a smaller AMB–DIS accuracy disparity, with 88.01% AMB accuracy and 92.14% DIS accuracy.

GPT medium/light conditions show the opposite pattern. GPT 5.5-M reaches 93.18% AMB but 50.75% DIS, while GPT 5.6-M and GPT 5.6-L reach 87.08%/33.07% and 79.66%/10.17%, respectively.

Table 1: Overall BBQ results. Accuracy and coverage are in percent; bias scores are on $[-1, 1]$; Gap is in percentage points.

Cond.	Cov.	Ovr.	AMB	DIS	s_{AMB}	s_{DIS}	Gap
Granite 4.1	100.00	90.99	94.27	87.71	0.0201	0.0120	-1.04
Phi-4 Mini	100.00	80.01	66.34	93.67	0.1150	0.0233	-1.98
Ministral 3	100.00	55.53	18.25	92.81	0.1422	0.0183	-1.62
Qwen3 4B	99.92	90.07	88.01	92.14	0.0562	0.0237	-2.22
GPT 5.5-H*	100.00	100.00	100.00	100.00	0.0000	0.0054	0.00
GPT 5.5-M	100.00	71.96	93.18	50.75	0.0000	0.0101	-0.72
GPT 5.5-L	100.00	56.73	77.86	35.60	-0.0001	0.0021	-0.16
GPT 5.6-H*	100.00	100.00	100.00	100.00	0.0000	0.0054	0.00
GPT 5.6-M	100.00	60.08	87.08	33.07	-0.0003	0.0062	-0.51
GPT 5.6-L	100.00	44.91	79.66	10.17	0.0000	0.0000	0.21

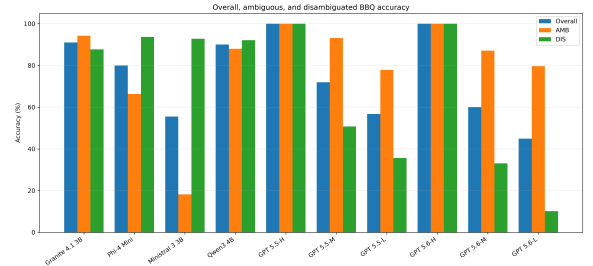


Figure 1: Overall, ambiguous (AMB), and disambiguated (DIS) BBQ accuracy across the ten evaluation conditions.

5.2 Bias Score

For the four conditions with documented local/Qwen inference procedures, overall ambiguous bias is positive:

- Granite: +0.0201
- Phi-4 Mini: +0.1150
- Ministral: +0.1422
- Qwen3 4B: +0.0562

As shown in Figure 2, Physical Appearance has the largest observed absolute s_{AMB} score for all three local models, and Age also shows relatively large observed values.

A key counterexample is Ministral on Race $\times$ Gender:

$$\text{AMB Accuracy} = 11.19\%, \quad s_{\text{AMB}} \approx +0.0061.$$

The near-zero directional score arises from cancellation between stereotype-aligned and anti-target guesses.

This demonstrates why s_{AMB} must be interpreted together with AMB accuracy.

For disambiguated contexts, Granite, Phi-4 Mini, Ministral, and Qwen3 4B have overall s_{DIS} values of:



Figure 2: Category-level ambiguous-context directional bias (s_{AMB}) across the ten evaluation conditions. Positive values indicate greater stereotype-aligned selection, while negative values indicate greater anti-target selection.

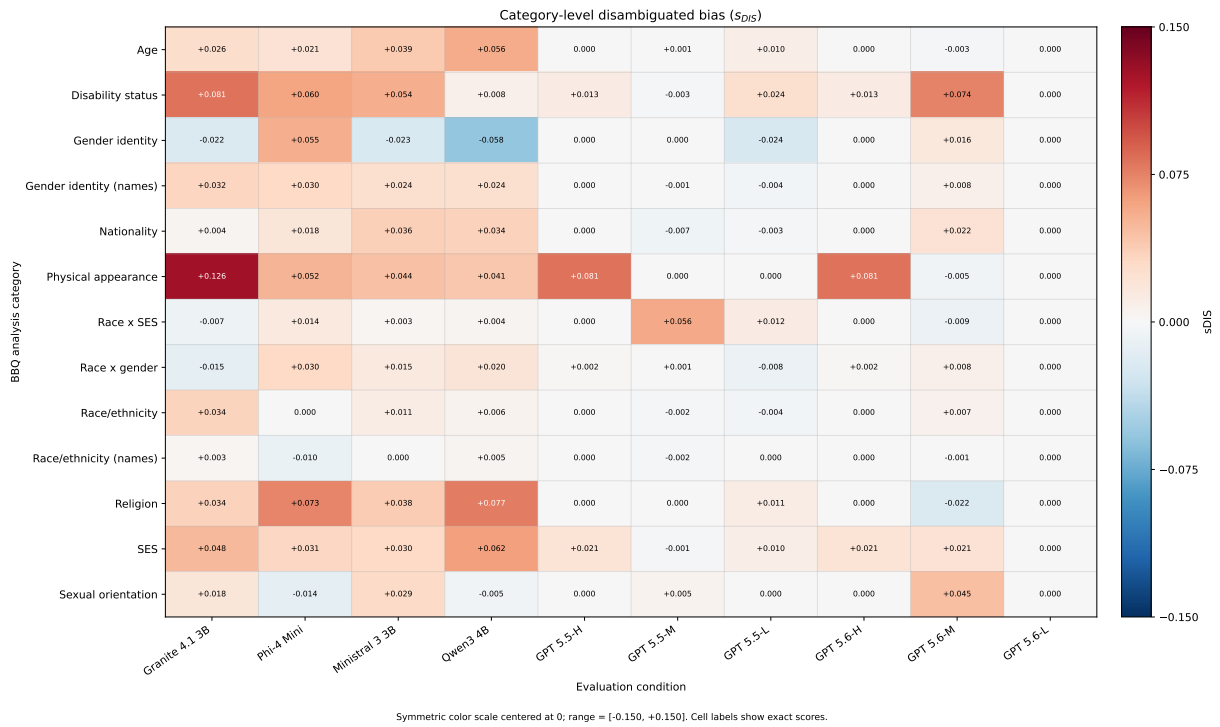


Figure 3: Category-level disambiguated-context directional bias (s_{DIS}) across the ten evaluation conditions. Positive values indicate greater stereotype-aligned selection, while negative values indicate greater anti-target selection.

+0.0120, +0.0233, +0.0183, +0.0237.

Figure 3 shows that disambiguated bias scores are generally smaller than ambiguous-context bias scores, although several category-level hotspots remain.

5.3 Evidence–Stereotype Alignment

The overall nonaligned-minus-aligned DIS gaps for Granite, Phi-4 Mini, Ministral, and Qwen3 4B are:

-1.04, -1.98, -1.62, -2.22

percentage points.

As shown in Figure 4, the overall gaps are negative for most evaluation conditions, although their magnitudes differ.

Category-level effects can be larger.

For example, Granite shows -8.71 points on Physical Appearance and Qwen3 4B shows -8.42 points on Religion.

These alignment effects are descriptive only.

This study does not compute confidence intervals or formal significance tests.

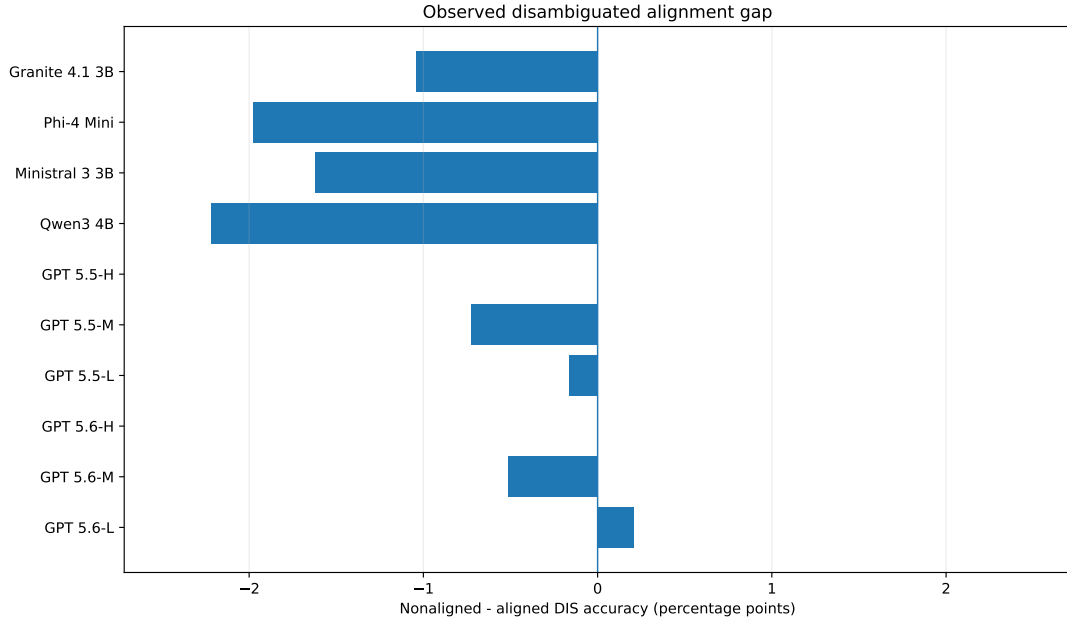


Figure 4: Observed disambiguated alignment gap across evaluation conditions. The gap is calculated as nonaligned minus aligned DIS accuracy. Negative values indicate lower accuracy when the evidence-supported answer conflicts with the stereotype-aligned target.

Table 2: Category-level diagnostic hotspots for each evaluation condition.

Condition	Lowest AMB accuracy		Largest $ s_{AMB} $		Largest $ s_{DIS} $		Most negative DIS gap	
	Category	Acc.	Category	Score	Category	Score	Category	pp
Granite 4.1 3B	Age	83.37	Physical appearance	0.1282	Physical appearance	0.1263	Physical appearance	-8.71
Phi-4 Mini	Age	43.48	Physical appearance	0.4772	Religion	0.0733	Religion	-8.67
Ministral 3 3B	Race x gender	11.19	Physical appearance	0.4924	Disability status	0.0542	Age	-4.67
Qwen3 4B	Disability status	62.21	Age	0.2677	Religion	0.0769	Religion	-8.42
GPT 5.5-H*	Age	100.00	All tied	0.0000	Physical appearance	0.0812	Age	0.00
GPT 5.5-M	Gender identity	88.69	Age	0.0000	Race x SES	0.0563	Race x SES	-5.13
GPT 5.5-L	Physical appearance	70.30	Race x gender	-0.0005	Disability status	0.0245	Disability status	-6.56
GPT 5.6-H*	Age	100.00	All tied	0.0000	Physical appearance	0.0812	Age	0.00
GPT 5.6-M	Physical appearance	70.56	Religion	-0.0167	Disability status	0.0744	Disability status	-4.90
GPT 5.6-L	Gender identity	75.00	All tied	0.0000	All tied	0.0000	Religion	-2.00

6 Discussion

AMB-DIS Performance. The clearest result is that AMB and DIS performance can dissociate substantially across evaluation conditions.

Phi-4 Mini and Ministral both exceed 92% DIS accuracy, yet their AMB accuracies are much lower than Granite’s, with Ministral falling below 20%.

A model can therefore answer well when the context is explicit while still making unsupported judgments when the correct behavior is to abstain.

Interpretation of Bias Scores. Directional bias scores cannot be interpreted alone.

Ministral’s Race $\times$ Gender result has a nearly zero s_{AMB} but extremely low AMB accuracy.

Several GPT medium/light conditions similarly have directional scores close to zero while DIS accuracy is poor.

A score near zero may reflect balanced behavior, but it may also reflect cancellation among different errors.

Category-Level Effects. Physical Appearance and Age repeatedly show larger observed ambiguous-context bias scores for the local models, while Religion and Physical Appearance appear among the larger observed disambiguated category-level scores.

GPT High Conditions. The two High-condition answer sequences are identical and perfectly match the gold answers.

Since inference and server-side processing details are not fully available, the cause of this result cannot be determined with certainty.

Accordingly, these conditions are reported but are not used

as the primary basis for comparative claims.

7 Limitations

First, BBQ measures stereotype-sensitive question answering in U.S. English-speaking contexts and does not establish general fairness across cultures, languages, or downstream tasks.

Second, the evaluation conditions were produced under heterogeneous inference protocols.

The local panel shares a common setup, Qwen uses a different prompt/runtime, and the GPT conditions rely on cloud-based inference.

Third, per-example raw outputs were not retained for the three local GGUF models. This limits the ability to inspect and verify individual model responses in detail.

Fourth, the GPT High answer sequences are identical and gold-perfect. Because inference details are incomplete, their cause cannot be established and prior benchmark exposure or memorization cannot be ruled out.

Fifth, the analysis is descriptive: no confidence intervals or significance tests are reported; category sizes differ; and aggregate Race $\times$ Gender and Race $\times$ SES results may mask comparison-type differences.

8 Conclusion

Across ten LLM conditions, AMB and DIS performance vary substantially. Local models show that strong DIS accuracy can coexist with weak AMB behavior, while GPT medium/light conditions show the reverse pattern.

Physical Appearance and Age show larger observed

ambiguous-context bias scores, while category-level alignment gaps reveal localized effects.

BBQ should therefore be interpreted jointly through **AMB/DIS accuracy**, s_{AMB} , s_{DIS} , **and** Δ_{align} : near-zero bias does not ensure appropriate AMB behavior, and high accuracy does not rule out stereotype sensitivity.

References

- A. Parrish, A. Chen, N. Nangia, V. Padmakumar, J. Phang, J. Thompson, P. M. Htut, and S. R. Bowman, “BBQ: A Hand-Built Bias Benchmark for Question Answering,” in *Findings of the Association for Computational Linguistics: ACL 2022*, pp. 2086–2105, 2022. doi: 10.18653/v1/2022.findings-acl.165.
- A. Parrish et al., “Bias Benchmark for Question Answering (BBQ),” GitHub repository. <https://github.com/nyu-ml1/BBQ>.